



University Institute of Engineering
Department of Computer Science & Engineering

Experiment-9

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Branch: BE CSE (cyber security)
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Subject Name: ADBMS

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1. Aim of the practical: To study and implement **PostgreSQL Stored Procedures** using input and output parameters, conditional logic, database operations, and RAISE NOTICE messages to process employee and product-related data..

2. Questions

(A) An organization stores employee details in the employee_info table. Create a PostgreSQL stored procedure that accepts an employee's **gender** as an input parameter and calculates the total number of employees belonging to that gender. The calculated count should:

- Be returned through an **OUT parameter**.
- Also be displayed as a message using RAISE NOTICE.

For example, if Male is passed to the procedure, it should count all employees whose gender is Male.

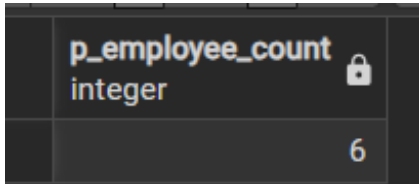
employee_info

Column Name	Data Type	Description
id	SERIAL	Unique employee ID
name	VARCHAR(50)	Employee name
gender	VARCHAR(10)	Gender of employee
salary	NUMERIC(10,2)	Employee salary
city	VARCHAR(50)	Employee city

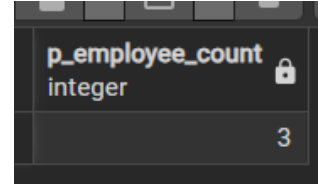
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Solution:



p_employee_count	integer	6
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p_employee_count	integer	3
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(B) An e-commerce company maintains product stock information in the products table and completed orders in the sales table.

Create a PostgreSQL stored procedure that accepts a **product name** and **required quantity** as input parameters and processes the purchase only when sufficient stock is available.

If sufficient stock exists, the procedure should:

- Retrieve the product code and price.
- Calculate the total sale amount.
- Insert a new order into the sales table.
- Reduce the available stock.
- Increase the total quantity sold.
- Display a success message.

If sufficient stock is not available, the procedure should:

- Not insert or update any record.
- Display an appropriate insufficient-quantity message.

products

Column Name	Data Type	Description
product_code	VARCHAR(10)	Unique product code
product_name	VARCHAR(100)	Product name
price	NUMERIC(10,2)	Price of one unit
quantity_remaining	INT	Current available stock
quantity_sold	INT	Total quantity sold

sale

Column Name	Data Type	Description
order_id	SERIAL	Unique order ID
order_date	DATE	Date on which order was placed
product_code	VARCHAR(10)	Product purchased
quantity_order ed	INT	Quantity purchased
sale_price	NUMERIC(10,2)	Total order amount



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Solution:

Data Output Messages Notifications

NOTICE: PRODUCT SOLD...! Order placed successfully for 1 unit(s) of MacBook Pro 14".
CALL

Query returned successfully in 53 msec.

3. Learning Outcomes:

- Learned to create and execute stored procedures in PostgreSQL.
- Applied input and output parameters in stored procedures.
- Used conditional logic to process database operations.
- Learned to update and manage product and sales data through procedures.
- Developed practical skills in using RAISE NOTICE for database messages.